Satellite Constellation Design

Optimal Constellation Design for Space-Based Wireless Power Transfer

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Abstract

This project builds a complete constellation-design pipeline that answers a concrete engineering question: where should ground stations be placed, which orbits should satellites fly, and what is the minimum number of satellites that keeps every station continuously powered?

The system places 50 ground rectennas by segmenting an Earth land/ocean map with K-Means and Voronoi partitioning, propagates a grid of candidate orbits, computes satellite-to-rectenna visibility fully vectorized on the GPU, and then selects the smallest viable constellation with a mixed-integer linear program. A from-scratch spacecraft dynamics model underpins the physics, and the whole solution is visualized in NVIDIA Isaac Sim.

Technical Highlights

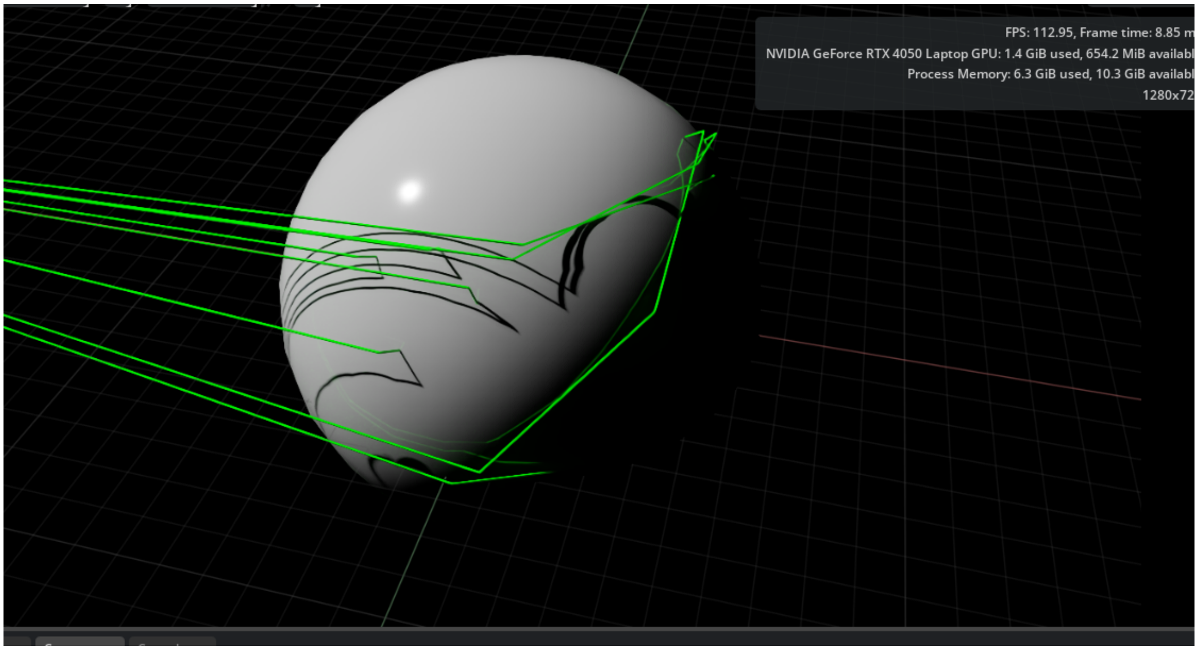
- **Data-driven station placement:** 50 ground rectennas positioned via K-Means + Voronoi segmentation of a global land/ocean map.
- **GPU-vectorized visibility:** satellite-to-rectenna access computed across a (1183 × 50 × time) tensor entirely on the GPU with PyTorch/CUDA.
- **Optimal sizing:** a mixed-integer linear program (PuLP/CBC) selects the minimum constellation subject to per-station coverage constraints.
- **Independent verification:** a separate verifier re-checks every coverage constraint on the chosen solution.
- **First-principles dynamics:** a 12-DOF rigid-body spacecraft model derived in both Newton–Euler and Lagrangian formulations, cross-validated against MuJoCo.
- **Immersive visualization:** the constellation and animated multi-hop power beams rendered in NVIDIA Isaac Sim.

Technologies Used

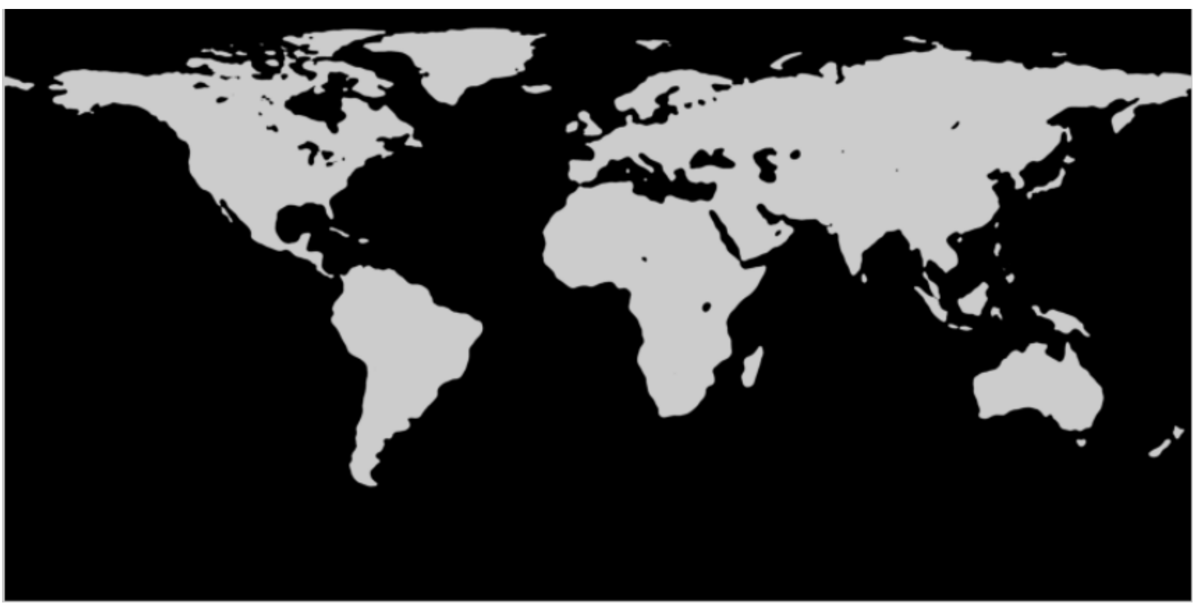
- **Languages:** Python
- **Numerical / GPU:** PyTorch (CUDA), NumPy

- **Optimization:** Mixed-Integer Linear Programming with PuLP / CBC
- **Geospatial / CV:** K-Means & spherical Voronoi (scikit-learn), OpenCV map processing
- **Dynamics:** full 12-DOF rigid-body model in Newton–Euler & Lagrangian formulations, cross-validated against MuJoCo
- **Visualization:** NVIDIA Isaac Sim (USD scene graph, animated multi-hop power beams)

Figures

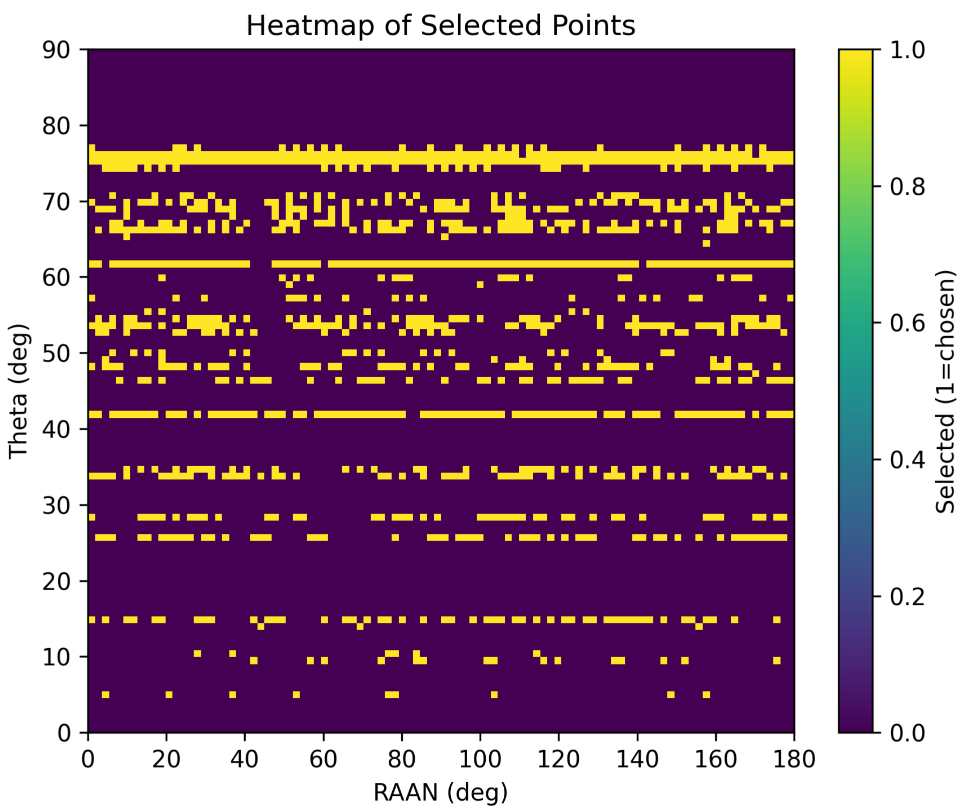


Constellation and multi-hop power beams visualized in NVIDIA Isaac Sim.

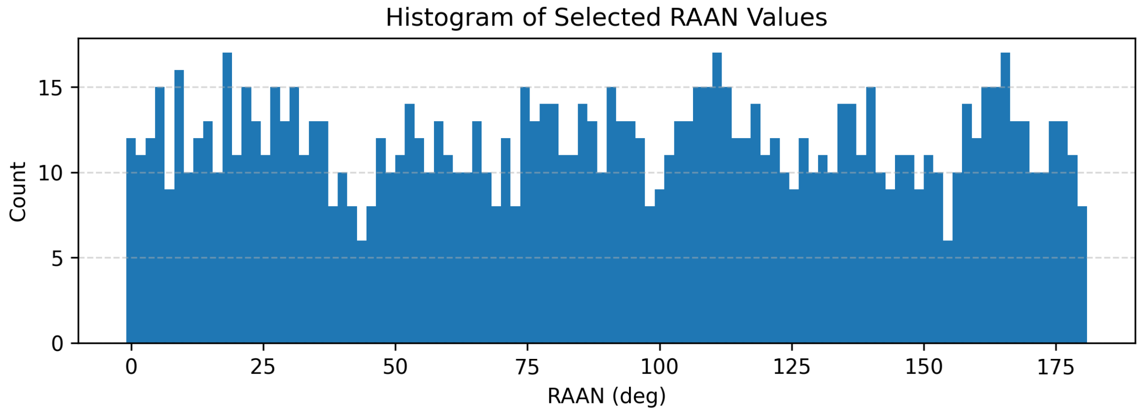


Global land/ocean map used as input for ground-station placement.

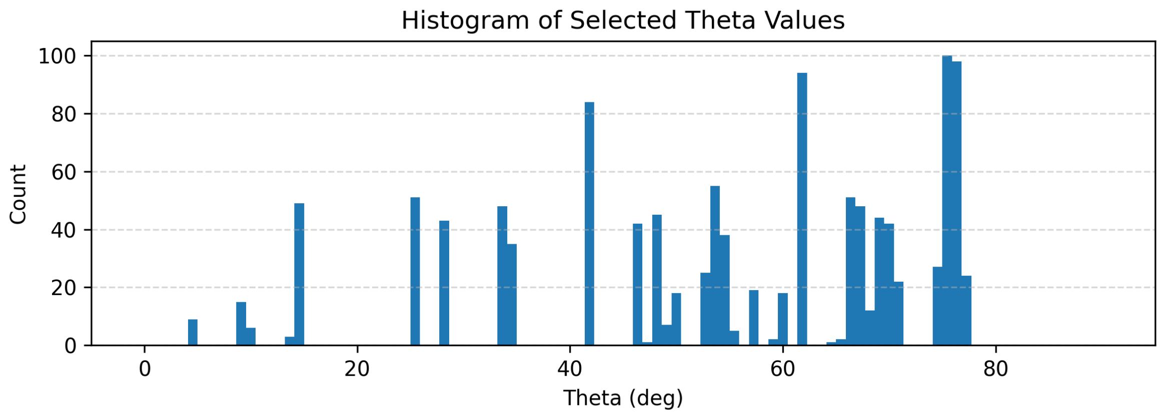
K-Means + Voronoi segmentation of the Earth map into candidate service regions.



Heatmap of selected ground-rectenna locations.



Distribution of right ascension of ascending node (RAAN) across candidate orbits.



Distribution of orbital true-anomaly / phase across the candidate grid.

Outcome

The pipeline delivers a minimal, fully-verified satellite constellation that keeps all 50 ground stations powered, combining data-driven placement, GPU-scale visibility computation, and provably-constrained optimization into one coherent design tool.